

backend/rag/keyword_index.py Documentation (Roman Urdu)

Yeh script search engine ka purana magar ahem hissa hy (Lexical Search).

File Ka Maqsad (Purpose)

Vectors/Embeddings kabhi kbhi "Exact Match" miss kr dytey hain. Misal ke toor par agr user ny specifically ID "A-1029" search ki hy, tu vector model ussy general similar ids dedy ga. Lekin ye file SQLite FTS5 (Full-Text Search) index k istmzl krh BM25 Algorithm (Google ka puraana search logic) chlty ai takh Exact Words (Keywords) pkry jan skn!

Code Blocks aur Classes ki Tafseel

1. class KeywordIndex

- **Database Tables:** `chunks_fts` (jis mai search lgty ho) aur `chunks_meta` create karta ay.
- **search():** User ki query k alfaz par "Escape/Sanitize" process karty hz `_escape_fts_query` ta ky ghalat chars fail na hn. Phir SQL MATCH aur bm25 scoring lgty hs jo score list wapis bhjtee e.